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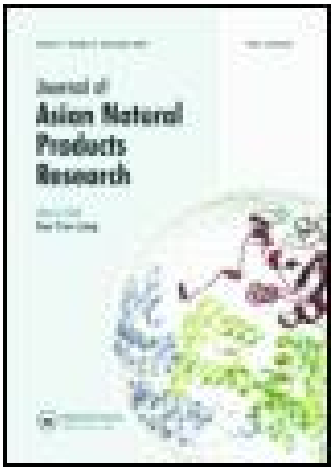

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Three new cyathane diterpenoids from the medicinal fungus Cyathus africanus

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Three new cyathane diterpenoids from the medicinal fungus *Cyathus africanus*

Jun-Jie Han[a], Lei Zhang[b], Jie-Kun Xu[c], Li Bao[a], Feng Zhao[d], Yu-Hui Chen[e], Wei-Ku Zhang[b]*[1] and Hong-Wei Liu[a]*[1]

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(*Received 7 January 2015; final version received 20 April 2015*)

Three new cyathane diterpenoids, cyathin W (**1**), cyathin V (**2**), and cyathin T (**3**), were isolated from the solid culture of *Cyathus africanus*. The structures and configurations of these new compounds were elucidated on the basis of comprehensive spectroscopic analysis including 1D NMR, 2D NMR (HSQC, HMBC, NOESY), and HR-ESI-MS experiments. Compounds **1** and **3** showed moderate inhibition against nitric oxide production in lipopolysaccaride-activated macrophages with IC_{50} value of 80.07 and 88.87 μM, respectively. In cytotoxicity assay, compound **1** showed weak cytotoxicity against K562 cell line with IC_{50} value of 12.1 μM.

Keywords: Nidulariaceae; cyathane diterpenoids; *Cyathus africanus*; NO inhibition; cytotoxicity

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1. Introduction

Fungi are one of the main sources of natural medicines because of their abundant and unique metabolites [1]. Medicinal fungi are distributed throughout the world with about 2500 species, and are used in Chinese traditional medicine historically [2]. In particularly, *Cyathus* is a genus of medical fungi which belongs to the family Nidulariaceae. Many *Cyathus* species are noted for their prolific producers of bioactive cyathane diterpenoids with the presence of a characteristic 5-6-7 tricarbocyclic fused core structure [3]. In 1972, Ayer and coworkers first reported the isolation of cyathin A_3 and allocyathin B_3 from the liquid culture of *Cyathus helenae* [4]. Numerous other cyathane diterpenoids were discovered following this report from a diverse variety of fungi, sponges, and fruiting plants [5–24]. Cyathane diterpenoids had been found to exhibit antimicrobial, antibiotic, anti-inflammatory, antibacterial, anti-proliferative, apoptosis-inducible, cytotoxic activities, and stimulating activity of nerve growth factor synthesis activities [5–19]. In our ongoing search for more cyathane diterpenoids, the NO inhibitory and cytotoxic compounds from the fungus *Cyathus africanus* have been investigated [6]. In this study, three new cyathane diterpenoids (Figure 1) were isolated from the EtOAc extract of the solid culture of the fungus *C. africanus*. Structures of compounds **1**–**3** were established on the basis of spectroscopic methods. Details of the isolation, structural elucidation, and bioactivity evaluation of compounds **1**–**3** are presented herein.

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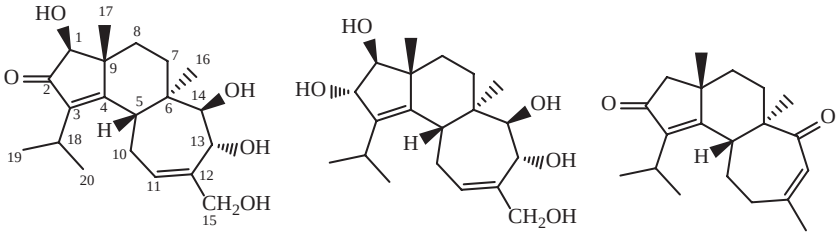

cyathinW (**1**) cyathinV (**2**) cyathinT (**3**)

Figure 1. Structures of compounds **1**–**3**.

2. Results and discussion

Compound **1** was isolated as white powder. The molecular formula of **1** was determined to be $C_{20}H_{30}O_5$ (six degrees of unsaturation) by the HR-ESI-MS ion at *m/z* 373.1993 $[M + Na]^+$. The ^{1}H NMR (Table 1) spectrum revealed the presence of four methyls [δ_H 1.02 (3H, s, H_3-16), 1.16 (3H, $J = 6.9$ Hz, H_3-20), 1.26 (3H, d, $J = 6.9$ Hz, H_3-19), 1.28 (3H, s, H_3-17)], one olefinic proton [δ_H 5.90 (1H, d, $J = 8.1$ Hz, H-11)], three oxymethines [δ_H 3.49 (1H, d, $J = 9.3$ Hz, H-14), 3.63 (1H, s, H-1), 4.56 (1H, d, $J = 9.3$ Hz, H-13)], and two oxymethylene protons [δ_H 4.10 (1H, d, $J = 12.8$ Hz, H-15β), 4.26

Table 1. ^{1}H and ^{13}C NMR spectral data of compounds **1**–**3**.

	1[a]		**2**[a]		**3**[a]	
No	δ_H (*J* in Hz)	δ_C	δ_H (*J* in Hz)	δ_C	δ_H (*J* in Hz)	δ_C
1	3.63 (s)	83.8	3.48 (d, 6.0)	82.1	2.19 (o[b])	53.0
	–	–	–	–	2.19 (o[b])	–
2	–	209.8	4.43 (d, 6.0)	75.3	–	210.6
3	–	141.9	–	140.7	–	144.4
4	–	176.4	–	146.7	–	179.6
5	2.93 (o[b])	47.0	2.55 (o[b])	45.8	3.24 (d, 8.7)	45.6
6	–	45.2	–	43.4	–	58.2
7	1.64 (o[b])	29.1	1.59 (dd, 4.4, 13.6)	28.9	1.45 (o[b])	35.5
	1.85 (o[b])	–	1.65 (o[b])	–	1.85 (o[b])	–
8	1.51 (o[b])	37.4	1.39 (td, 6.0, 13.0)	38.1	1.57 (o[b])	39.0
	1.64 (o[b])	–	1.76 (o[b])	–	1.85 (o[b])	–
9	–	47.9	–	50.9	–	43.6
10	2.34 (dd, 8.4, 16.9)	28.4	2.45 (dd, 7.9, 14.6)	29.2	2.07 (o[b])	27.0
	2.75 (o[b])	–	2.51 (o[b])	–	2.28 (o[b])	–
11	5.90 (d, 8.1)	128.8	5.87 (d, 8.0)	129.8	2.48 (o[b])	39.2
	–		–	–	2.56 (o[b])	–
12	–	143.8	–	143.3	–	156.5
13	4.56 (d, 9.3)	69.3	4.51 (d, 9.5)	69.1	5.92 (s)	127.3
14	3.49 (d, 9.3)	80.8	3.45 (d, 9.5)	81.3	–	209.2
15	4.10 (d, 12.8)	66.5	4.08 (d, 6.0)	66.6	1.96 (s)	26.9
	4.26 (d, 12.8)	–	4.25 (d, 6.0)	–	–	–
16	1.02 (s)	18.0	0.96 (s)	18.0	1.13 (s)	16.4
17	1.28 (s)	20.4	1.08 (s)	20.6	1.28 (s)	23.9
18	2.90 (sept, 6.9)	27.2	2.92 (sept, 6.8)	28.4	2.96 (sept, 6.8)	27.2
19	1.26 (d, 6.9)	20.0	1.06 (d, 6.8)	22.5	1.22 (d, 6.8)	19.8
20	1.16 (d, 6.9)	21.6	1.15 (d, 6.8)	23.2	1.21 (d, 6.8)	21.2

[a] ^{1}H NMR at 500 Hz, ^{13}C NMR at 125 Hz in CD_3OD.
[b] Overlapped peaks.

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(1H, d, $J = 12.8$ Hz, H-15α)]. The ^{13}C NMR spectrum (Table 1) showed 20 carbons including four methyls, four methylenes (one oxygenated), five methines (three oxygenated), two quaternary carbons, four sp^2 carbons at δ_C 128.8, 141.9, 143.8, and 176.4 and one carbonyl carbon. In the HMBC experiment (Figure 2), correlations from H_3-19 (δ_H 1.26) to C-3, from H_3-20 (δ_H 1.16) to C-3, and from H-18 (δ_H2.90) to C-2 placed an isopropyl moiety at C-3. Besides, H-5 (δ_H 2.93) showed long-range correlations with C-3, C-4, and C-6, H-15 (δ_H 4.10, 4.26) showed long-range correlations with C-11, C-12, and C-13, H_3-16 (δ_H 1.02) showed long-range correlations with C-5, C-6, C-7, and C-14, and H_3-17 (δ_H 1.28) showed long-range correlations with C-1, C-4, C-8, and C-9. Interpretation of the above ^{1}H NMR, ^{13}C NMR, and HMBC spectral data and comparison with cyathin M [8] demonstrated that compound **1** has a cyathane-type skeleton and has the same planar structure as cyathin M. The only difference between **1** and cyathin M was in the stereoscopic configuration at C-1. The NOESY correlations of H-8α with H-1, H_3-16, and H-14, and H-7β with H_3-17, H-5, and H-13 revealed the α orientation of H-1, H_3-16, and H-14, and the β orientation of H_3-17, H-5, and H-13 (Figure 3). The CD spectrum (Figure 4) of **1** showed a negative Cotton effect at 250 nm, which is consistent with the axiom of an α,β-unsaturated cycloketone [20] (Figure 5). Therefore, the absolute configuration at C-1 of cyathin W was determined as S. However, in the CD spectrum of cyathin M, a strong positive Cotton effect was present at 250 nm (Figure 4), which was consistent with the axiom of an α,β-unsaturated cycloketone [20] (Figure 5). The absolute configuration at C-1 of cyathin M was established as *R*. On the basis of the above data, the structure of **1** was assigned and designated as cyathin W.

Compound **2**, obtained as white powder, possessed a molecular formula of $C_{20}H_{32}O_5$ (five degrees of unsaturation) at *m/z* 375.2139 $[M + Na]^+$ by HR-ESI-MS. The ^{1}H and ^{13}C NMR spectroscopic data (Table 1) of **2** showed the presence of a cyathane skeleton, which indicated its structure is similar to that of **1** except for C-2. Interpretation of the ^{1}H NMR, ^{13}C NMR, and HSQC spectral data indicated the presence of four oxygenated methines [δ_H 3.48 (1H, $J = 6.0$ Hz), δc 82.1; 4.43 (1H, $J = 6.0$ Hz), δc 75.3; 4.51 (1H, $J = 9.5$ Hz), δc 69.1; 3.45 (1H, $J = 9.5$ Hz), δc 81.3]. In the HMBC spectrum, correlations from H-1 (δ_H 3.48) to C-2 and C-9, from H-2 (δ_H 4.43) to C-1, C-3, and C-4, from H_3-19 (δ_H 1.06) to C-3 and C-18, from H_3-20 (δ_H 1.15) to C-3, from H-18 (δ_H 2.92) to C-2, from H-5 (δ_H 2.55) to C-3, C-4, C-6, and C-10, from H-15 (δ_H 4.08, 4.25) to C-11, C-12, and C-13, from H-13 (δ_H 4.51) to C-12, C-14, and C-15, from H-14 (δ_H 3.45) to C-7 and C-13, from H_3-16 (δ_H 0.96) showed long-range correlations with C-5, C-6, C-7, and C-14, and from H_3-17 (δ_H 1.08) to

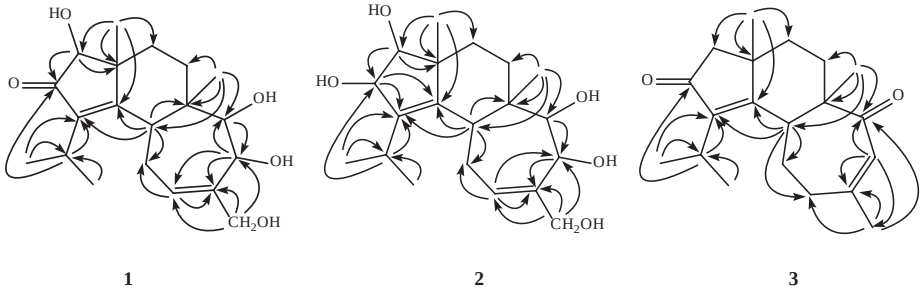

Figure 2. Key HMBC correlations of compounds **1**–**3**.

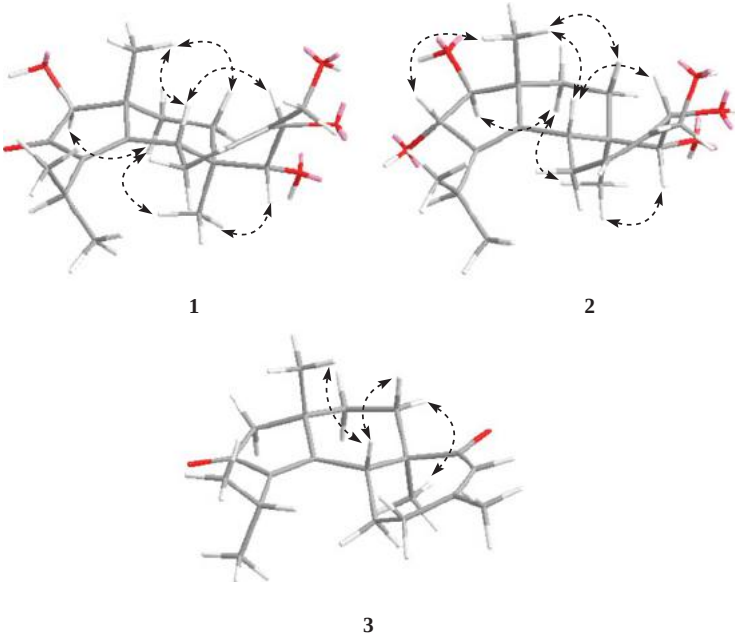

Figure 3. Key NOESY correlations of compounds **1**–**3**.

C-1, C-4, C-8, and C-9 supported the structure of **2** as described in Figure 1. The larger coupling constant ($J = 6.0$ Hz) between H-1 and H-2 supported the *trans* configuration. Likewise, the *trans* configuration between H-13 and H-14 was deduced from the large coupling constant of 9.5 Hz. The relative configuration of **2** was confirmed by NOESY experiment (Figure 3). H-1, H-14, and H_3-16 were in the α-orientation, and H-2, H-5, H-13, and H_3-17 were in the β-orientation due to NOESY cross-peaks of H_3-16/H-8α, H-14/ H3-16, H-1/ H3-8α, H3-17/ H-7β, H-2/ H3-17, H-5/ H_3-17, and H-13/H-5. From the above-mentioned evidence, compound **2** was determined and named as cyathin V.

Compound **3** had the molecular formula of $C_{20}H_{28}O_2$ (seven degrees of unsaturation) at *m/z* 301.2166 $[M + H]^+$ by HR-ESI-MS. The 1D (Table 1) and 2D NMR data of **3** clearly revealed that it shared the same cyathane skeleton as cyafrin B_4 [21,22], the major difference between **3** and cyafrin B_4 occurred in the cycloheptane ring. The signals at δ_H 2.48 (1H, o, H-11β), 2.56 (1H, o, H-11α), and δ_C 39.2 (C-11) and HMBC correlations (Figure 2) from H_3-15 to C-11 and H-5 to C-11 implied a methylene at C-11.

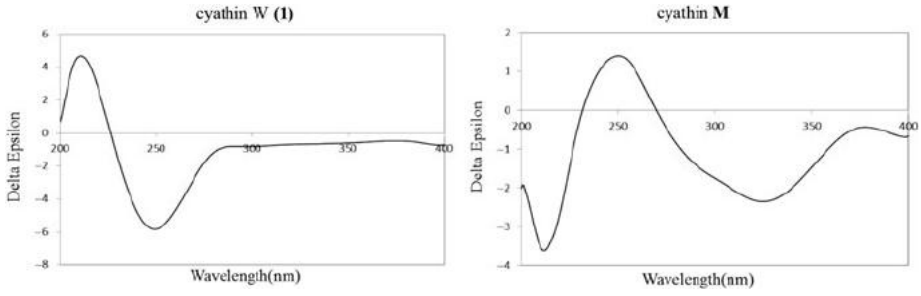

Figure 4. CD curves of cyathin W (**1**) and cyathin M.

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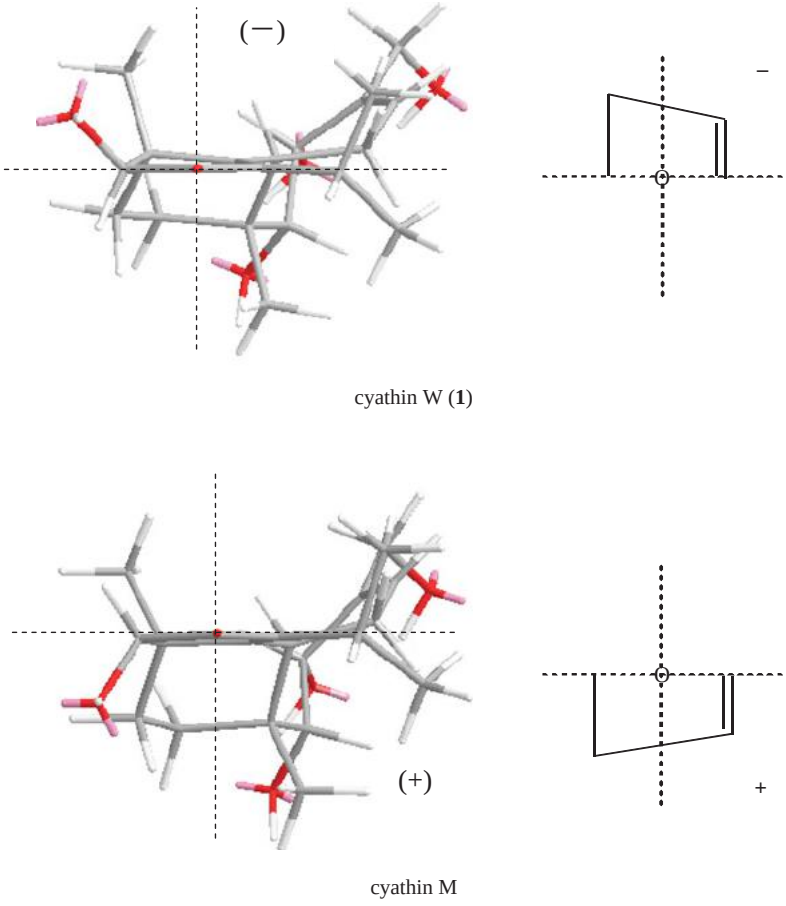

Figure 5. The Cotton effect of α, β-unsaturated cycloketone moiety of cyathin W (**1**) and cyathin M.

Likewise, the ^{1}H and ^{13}C NMR spectrum revealed the appearance of a methyl [δ_H 1.96 (3H, s); δ_C 26.9] and HMBC correlations (Figure 2) from H_3-15 to C-11, C-12, and C-14 implied the methyl placed at C-12. The relative configuration of **3** was assigned on the basis of NOESY data (Figure 3). Correlations of H_3-17 with H-7β and H-5, and H_3-16 with H-7α indicated β configurations for H-5, H_3-17 and α configuration for H_3-16. Thus, the structure of **3** was elucidated as cyathin T.

Because of the diverse bioactivities of cyathane diterpenoids [5–19], the nitric oxide inhibitory activity and cytotoxicity of compounds **1**–**3** were evaluated in this study. As shown in Table 2, compounds **1** and **3** showed moderate inhibition against nitric oxide production in lipopolysaccaride (LPS)-activated macrophages with an IC_{50} value of 80.07 and 88.87 μM, respectively. Hydrocortisone was used as the positive control with IC_{50} value of 53.80 μM. In cytotoxicity assay against two cell lines HeLa and K562 (Table 2), compound **1** showed weak cytotoxicity against K562 cell line with IC_{50} value of 12.1 μM.

3. Experimental

3.1. General experimental procedures

Optical rotations were measured on a Perkin-Elmer 241 polarimeter (Perkin-Elmer, Waltham, USA), and UV data

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Table 2. Bioactivities of compounds **1**–**3** (IC_{50}, μM).

Compounds	Anti-inflammation[a] No	Cytotoxicity Hela	K562
1	80.07	>100	12.1
2	>100	Inactive	96.5
3	88.87	Inactive	>100
Positive control	Hydrocortisone 53.80	5-FU 20.20	5-FU 6.82

[a] The cell growth was not influenced by compounds tested at 100 μM.

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were recorded on a Shimadzu Biospec-1601 spectrophotometer (Shimadzu, Kyoto, Japan). IR data were recorded using a Nicolet Magna-IR 750 spectrophotometer (Thermo Fisher Scientific Inc. Waltham, USA). CD spectra were recorded on a JASCO J-815 spectropolarimeter (JASCO Corporation, Tokyo, Japan), using CH_3OH as solvent. NMR data were acquired with Bruker Avance-500 spectrometer (Bruker Corporation, Bremen, Germany) using solvent signals (CD_3OD, δ_H3.30/δ_C49.9) as references. ESI-MS and HR-ESI-MS data were obtained using a Bruker APEX III 7.0 T spectrometer (Bruker Corporation, Bremen, Germany). TLC was carried out on Silica gel HSGF254 (Qingdao Haiyang Chemical Co., Ltd., Qingdao, China) and the spots were visualized by spraying with 10% H_2SO_4 and heating. Silica gel (200–300 mesh, Qingdao Haiyang Chemical Co., Ltd., Qingdao, China), Sephadex LH-20 (Amersham Biosciences, Piscataway, USA), and ODS (Lobar, 40–63 μm, Merck, Darmstadt, Germany) were used for column chromatography (CC). Preparative HPLC was performed on an Agilent 1200 HPLC system (Agilent Technologies Inc., Santa Clara, USA) using an YMC C8 reversed-phase column (YMC Co, Tokyo, Japan) (250 mm × 10 mm, 5 μm) with a flow rate of 2.5 ml/min.

3.2 Fungal material

The strain used in this work was isolated from the fruiting body of *C. africanus*, and identified by Prof. Yuhui Chen. The fungus was deposited in China General Microbiological Culture Collection (CGMCC 5.1117). The fungus was identified on the basis of the DNA sequences of the ITS1-5.8S-ITS2, ITS regions of their ribosomal RNA gene. The ITS gene sequence obtained in this study was submitted to GenBank and assigned accession number JX103204. Analysis of sequences showed 100% homology with that of fungus *C. africanus* in GenBank with the accession number of DQ463342.1. *C. africanus* was cultured on slants of potato dextrose agar at 25°C for 10 days. Agar plugs were inoculated in a 500 ml Erlenmeyer flask containing 120 ml of media (0.4% glucose, 1% malt extract, and 0.4% yeast extract; the final pH of the media was adjusted to 6.5) before sterilization, and incubated at 25°C on a rotary shaker at 170 rpm for 1 week. The scale-up fermentation was carried out in fifty 500 ml Fernbach flasks, each containing 80 g of rice and 120 ml of distilled water. Each flask was inoculated with 5.0 ml of the culture medium and incubated at 25°C for 40 days.

3.3 Exaction and isolation

The fermented rice substrate was extracted repeatedly with ethyl acetate by exhaustive maceration (3 × 5 L), and the organic solvent was evaporated to dryness under vacuum to afford the crude extract (25.5 g). The residue was dissolved in

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H_2O (400 ml) and partitioned with $CHCl_3$ (400 ml) to yield a $CHCl_3$-soluble fraction (17.8 g). The $CHCl_3$-soluble fraction (17.8 g) was subjected to a silica gel CC eluted with a gradient of *n*-hexane–ethyl acetate (*v/v*, 100:0, 100:1, 100:2, 100:5, 100:10, 100:15, 100:25, 100:35) and dichloromethane–acetone (*v/v*, 100:2, 100:4, 100:8, 100:15, 100:25, 100:35, 0:100) to give 11 fractions (CA-1–CA-11).

Fraction CA-4 (2.5 g) eluted with *n*-hexane–ethyl acetate (v/v, 100:15) was subjected to silica gel CC with a gradient of *n*-hexane–ethyl acetate to afford CA-4.1–CA-4.13. CA-4.3 (85 mg) was separated by semi-preparative HPLC, with acetonitrile–H_2O (55:45, v/v) as mobile phase to afford compound **1** (2.5 mg, $R_t = 35.0$ min). Fraction CA-10 (0.8 g) eluted with dichloromethane–acetone (v/v, 100:15) was subjected to silica gel CC with a gradient of dichloromethane–methanol to afford CA-10.1–CA-10.9. Compounds **2** (8.5 mg, $R_t = 25.0$ min) and **3** (4.5 mg, $R_t = 32.0$ min) were obtained from the subfraction CA-10.4 (82.1 mg) through preparative HPLC purification using 20% acetonitrile in water.

3.3.1 *Cyathin W (**1**)*

White powder (MeOH); $[\alpha]_D^{25}$ − 135 (*c* 0.32, MeOH); UV (MeOH) λ_{max} (log ε): 252 (3.53) nm; IR ν_{max}: 3358, 2935, 2874, 1692, 1459, 1203 cm^{-1}; CD ($c = 0.35 \times 10^{-3}$ M, MeOH): $\Delta\varepsilon_{215nm}$ + 14.92, $\Delta\varepsilon_{247nm}$ − 12.77, $\Delta\varepsilon_{284nm}$ + 0.72; ^1H NMR and ^{13}C NMR spectral data, see Table 1; positive HR-ESI-MS: *m/z* 373.1993 $[M + Na]^+$ (calcd for $C_{20}H_{30}O_5Na$, 373.1985).

3.3.2 *Cyathin V (**2**)*

White powder (MeOH); $[\alpha]_D^{25}$ − 86.3 (*c* 0.17, MeOH); UV (MeOH) λ_{max} (log ε): 252 (2.18) nm; CD ($c = 0.35 \times 10^{-3}$ M, MeOH): $\Delta\varepsilon_{214\,nm}$ + 0.48, $\Delta\varepsilon_{226\,nm}$ + 0.77, $\Delta\varepsilon_{329\,nm}$ − 0.22; IR ν_{max}: 3405, 2923, 2851, 1736, 1655, 1436, 1238 cm^{-1}; ^1H NMR and ^{13}C NMR spectral data, see Table 1; positive HR-ESI-MS: *m/z* 375.2139 $[M + Na]^+$ (calcd for $C_{20}H_{32}O_5Na$, 375.2142).

3.3.3 *Cyathin T (**3**)*

White powder ($CHCl_3$); $[\alpha]_D^{25}$ − 306 (*c* 0.33, MeOH); UV (MeOH) λ_{max} (log ε): 225 (3.98) nm; CD ($c = 0.42 \times 10^{-3}$ M, MeOH): $\Delta\varepsilon_{220\,nm}$ +2.67, $\Delta\varepsilon_{248\,nm}$ − 11.62, $\Delta\varepsilon_{335\,nm}$ +0.58; IR ν_{max}: 3359, 2956, 2871, 1697, 1641, 1456, 1208 cm^{-1}; ^1H NMR and ^{13}C NMR spectral data, see Table 1; positive HR-ESI-MS: *m/z* 301.2166 $[M + H]^+$ (calcd for $C_{20}H_{29}O_2$, 301.2162).

3.4 *Biological study*

3.4.1 *NO inhibition assay*

LPS, dimethyl sulfoxide (DMSO), 3-(4,5-dimethylthiazol-2-yl)-2,5-diphenyl-2H-tetrazolium bromide (MTT), and hydrocortisone were purchased from Sigma Co., Ltd, St. Louis, USA. Mouse monocyte-macrophages RAW 264.7 (ATCC TIB-71), RPMI 1640 medium, penicillin, streptomycin, and fetal bovine serum were from the Chinese Academy of Science and Invitrogen (Beijing, China), respectively. RAW 264.7 cells were maintained in RPMI 1640 medium supplemented with penicillin (100 U/ml), streptomycin (100 mg/ml), and 10% heat-inactivated fetal bovine serum at 37°C in a humidified incubator with 5% CO_2 and 95% air. The medium was routinely changed every 48 h. Compounds **1–3** were dissolved in DMSO and were further diluted with the culture medium to ensure a final concentration of DMSO below 0.2% in the assay. As this concentration of DMSO had no marked impact on the growth of the cell line tested, each well was filled with 200 ml cells (5×10^5 cells/ml). After 60 min incubation, the cells were treated with 1 mg/ml of LPS and various

concentrations of test compounds for 24 h. Control groups were added to equal volume of DMSO. As a parameter of NO release, the nitrite concentration was measured in the supernatant of RAW 264.7 cells by the Griess reagent system according to protocol. The inhibitory rate of the compounds on NO production induced by LPS was calculated by the NO_2^- levels [22,23]. The inhibition rate was calculated and plotted versus test concentrations to afford the IC_{50}. All assays were performed in triplicate.

3.4.2 Cytotoxicity bioassay

Cytotoxic activity for cancer cell lines including HeLa and K562 cell lines was evaluated with the MTT assay [24]. For bioassay, after treating cells with the tested compounds (DMSO as solvent) for 72 h, each well was filled with 50 μl of MTT/medium solution (0.5 mg/ml), and incubated at 37°C in a humidified atmosphere with 5% CO_2 air for 4 h. The MTT/medium was removed gently and replaced with DMSO (100 μl/well) to mix with the formazan crystals until dissolved. The bioactivity was detected at 540 nm by a microplate reader. The inhibition rate was computed and plotted versus test concentrations to afford the IC_{50}. 5-FU was used as a positive control that showed cytotoxicity against HeLa and K562 cell lines with IC_{50} of 20.20 and 6.82 μM, respectively.

Disclosure statement

No potential conflict of interest was reported by the authors.

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